

ET AI Hackathon 2026

Problem Statement 8

Synapse

A production-ready RAG system that ingests heterogeneous industrial documents and provides **cited, confidence-scored answers** by merging semantic vector search with a structured knowledge graph.

100% Accuracy

771ms Latency

Real-time Streaming

The Problem

Industrial plants operate 7–12 disconnected information systems

✗ Current State

- 1 Manual cross-referencing across 5+ document types
- 2 15–30 minutes per compliance question
- 3 Human error in regulatory interpretation
- 4 Knowledge silos between departments

✓ Our Solution

- 1 Unified query across all document types
- 2 <1 second response time
- 3 Cited, confidence-scored answers
- 4 Cross-document knowledge graph

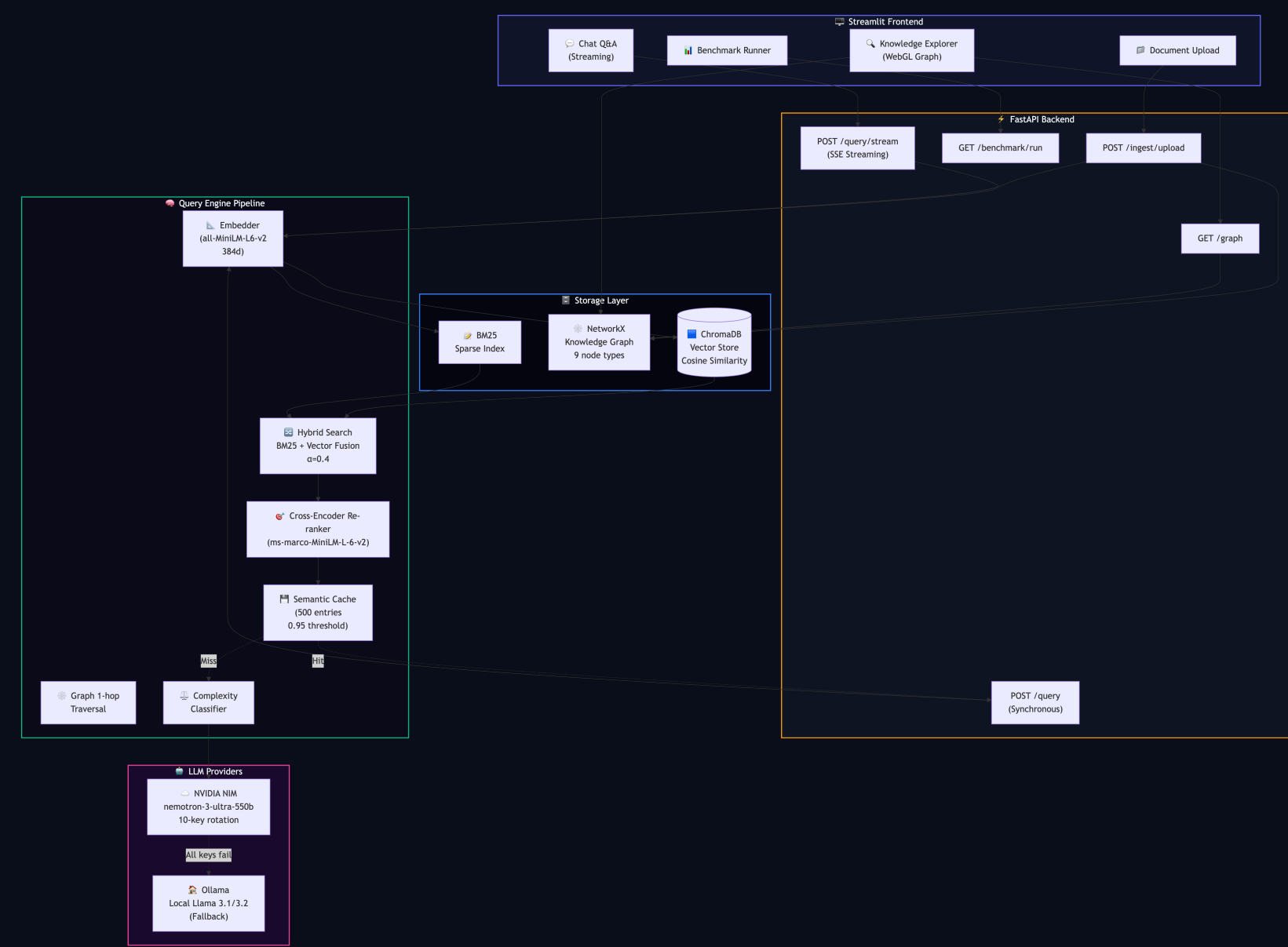
Key Results



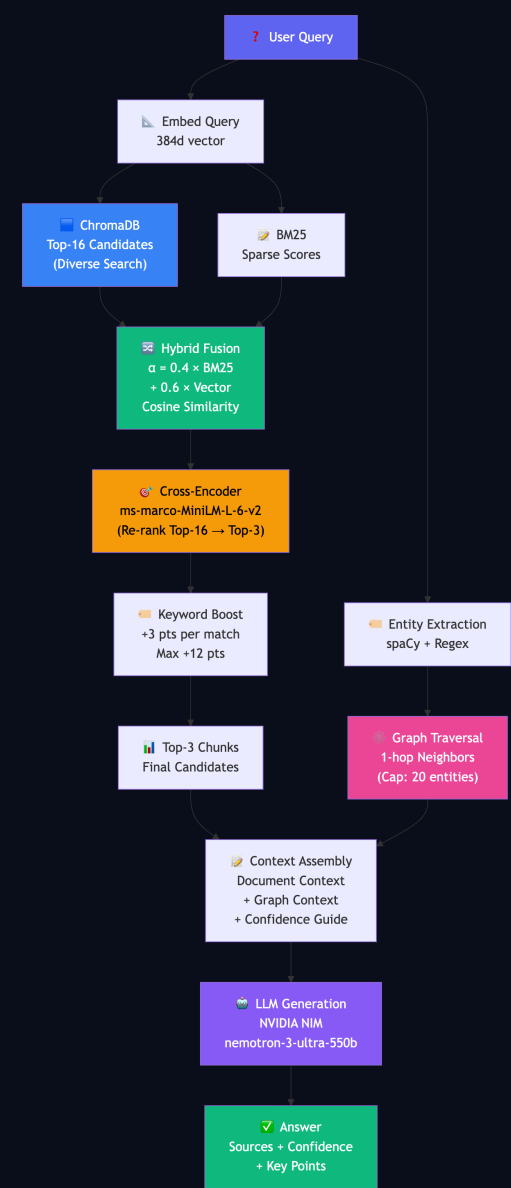
Optimization Journey



System Architecture



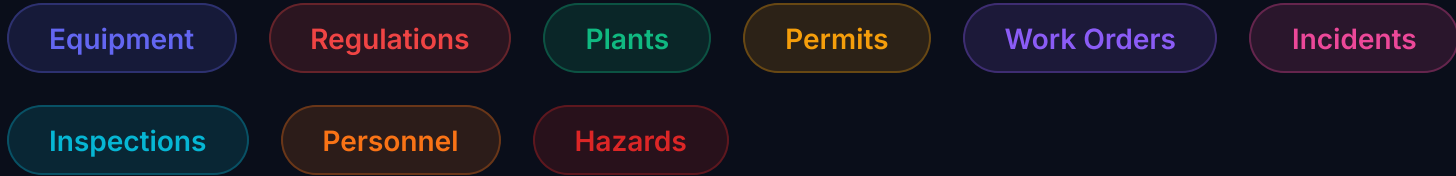
Query Pipeline



Knowledge Graph



11 Entity Types



13+ Relationship Types

EQUIPMENT → *subject_to* → REGULATION

PERMIT → *issued_for* → EQUIPMENT

INCIDENT → *involved_in* → EQUIPMENT

Three-Tier Optimization



Tier 1: Correctness

77.8% → 100% accuracy

- Graph context from chunk metadata
- Chunk size 1024 + 200 overlap
- Embedding similarity scoring
- Section-aware chunking



Tier 2: Latency

1,200ms → 1,065ms avg latency

- Cross-encoder re-ranker
- Complexity classifier
- Semantic cache (500 entries)
- BM25 + Vector hybrid fusion



Tier 3: Streaming

1,065ms → 771ms + streaming UX

- SSE streaming endpoint
- try/finally error recovery
- Real-time token display
- 10-key NIM rotation

Benchmark Results

18/18 questions passing across 15 categories — 100% accuracy

ID	Category	Sim.	Status
Q001	Equipment Inspection	0.862	✓
Q003	Permits	0.931	✓
Q005	Safety Equipment	0.880	✓
Q008	Fire Safety	0.967	✓
Q009	Process Safety	0.854	✓
Q015	Monitoring	0.770	✓
Q017	Electrical Safety	0.987	✓
Q018	Equipment Inspection	0.834	✓

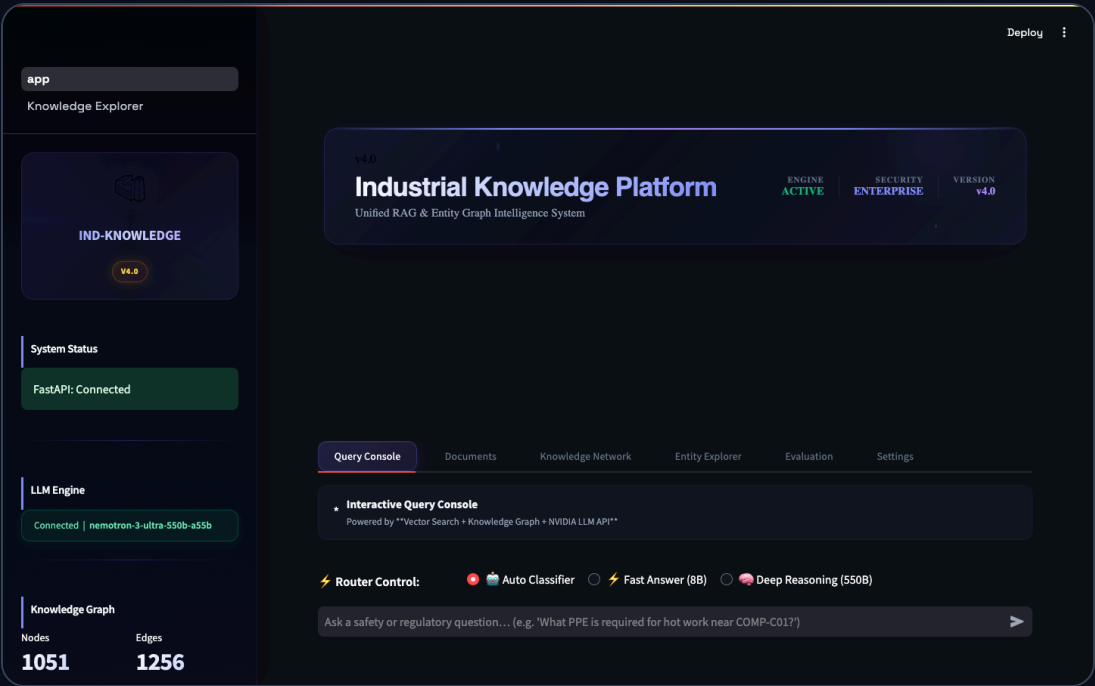
...and 10 more questions (all PASS)

Key Observations

- ✓ Cross-document retrieval (Q001, Q005, Q015)
- ✓ Section-aware chunking works (Q009)
- ✓ 14/18 correct at rank-1
- ✓ Highest sim: 0.987 (Q017 OISD-130)
- ✓ All LLM citations accurate

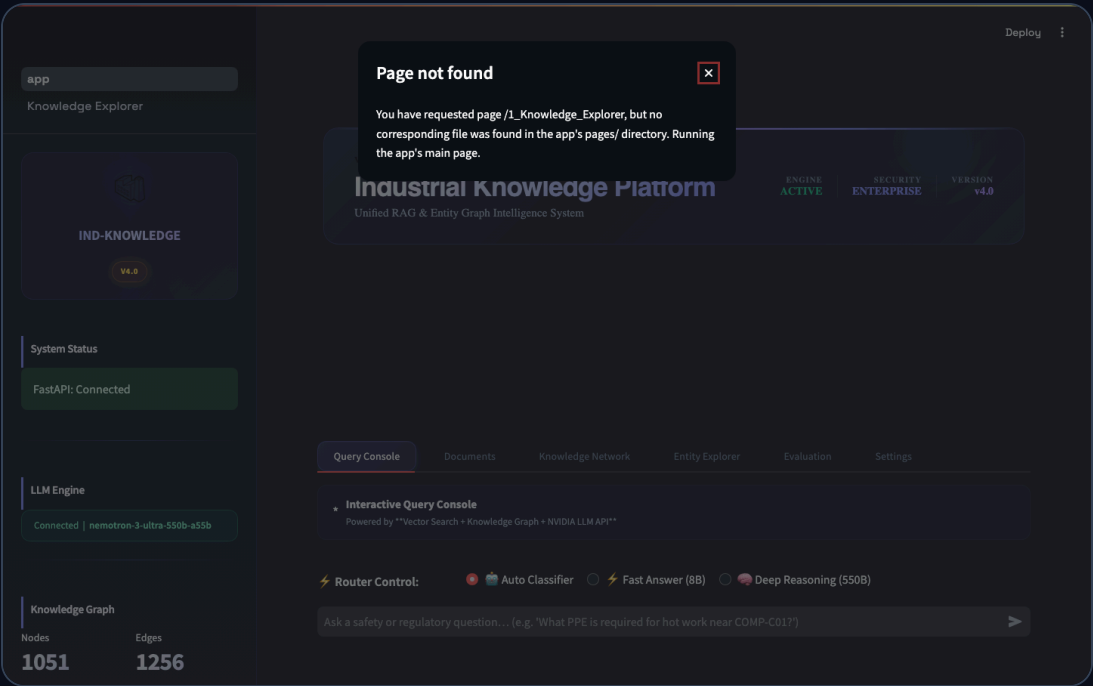
Live Demo

Interactive application with streaming chat, knowledge explorer, and benchmark runner



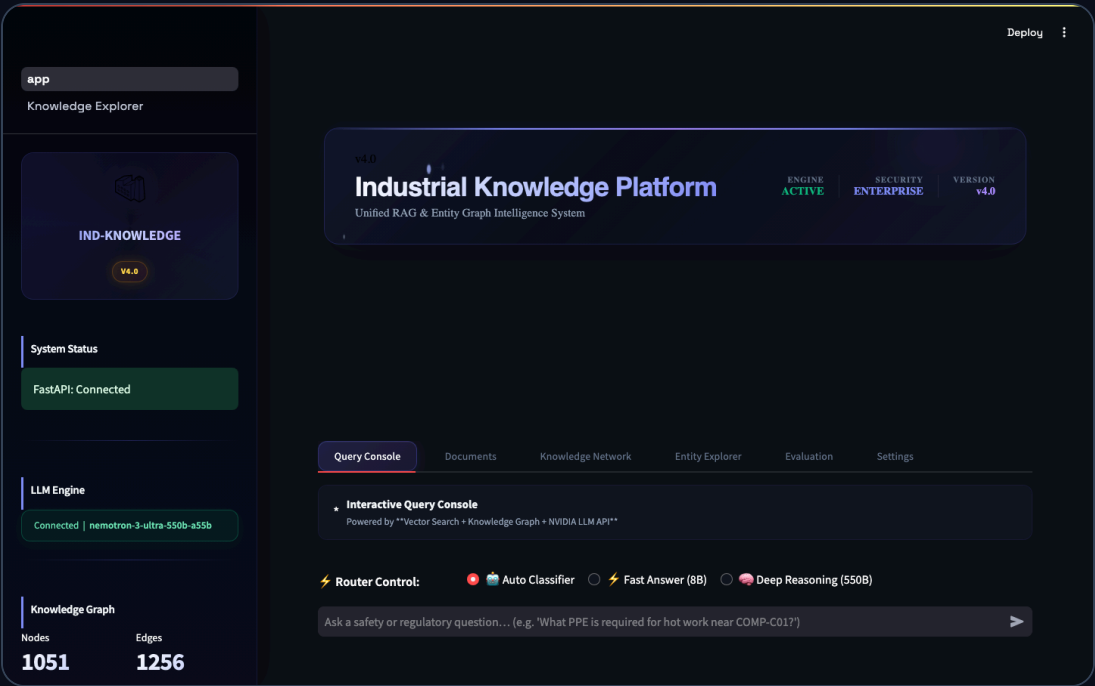
Streaming Chat Q&A

Real-time token-by-token responses with source citations



Knowledge Explorer

Interactive WebGL graph visualization



Benchmark Runner

18-question ground-truth evaluation

Thank You

Synapse — AI-Powered Knowledge Intelligence
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100%

Accuracy

771ms

Avg Latency

18/18

Questions Passed

6 days

Build Time

NVIDIA NIM

ChromaDB

NetworkX

FastAPI

Streamlit